

Submission Summary

Conference Name

2026 International Conference on Advancement of Cyber Security and Digital Forensics (ICACSDF)

Paper ID

264

Paper Title

Artificial Intelligence Driven Threat Detection in Modern Cybersecurity Systems

Abstract

The expansion of cloud services, distributed applications, and connected devices has made modern cybersecurity an increasingly data-intensive and adversarial problem. Traditional signature-based detectors remain effective against known attack patterns, yet they are poorly suited to rapidly evolving threats, including low-and-slow attacks, polymorphic malware, and previously unseen intrusions. This paper presents an Artificial Intelligence-driven threat detection framework that combines supervised machine learning with a hybrid anomaly-detection extension. Using the NSL-KDD benchmark dataset, seven algorithms are evaluated under both binary and multi-class classification settings. The strongest binary result is obtained by XGBoost, which reaches 80.08% accuracy and 96.78% precision, while Decision Tree performs best in the multi-class task with 76.80% accuracy. However, moderate recall indicates that a non-trivial fraction of attacks still evade detection. To address this gap, the paper proposes an extension that combines autoencoder reconstruction error, Isolation Forest anomaly scoring, and structural threat signals into a unified risk score. The analysis shows that hybrid AI is better aligned with modern cybersecurity requirements because it can balance false-alarm control with broader threat coverage.

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Primary Subject Area

AI & ML in Cybersecurity -> Network intrusion detection and mitigation

Secondary Subject Areas

Privacy and Information Security -> Database issues in privacy protection

Submission Files

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